

CS 736

COURSE PROJECT

**FISTA-Net: Learning A Fast Iterative
Shrinkage Thresholding Network
for Inverse Problems in Imaging**

Presented by - Team Tensor Titans

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OVERVIEW

- Problem statement and Motivation
- Important Terms
- Architecture of Fista-Net
- Experiment details
- Dataset
- Results
- Limitations
- Conclusion
- Future Plans
- References

PROBLEM STATEMENT AND MOTIVATION

- PS:

- Solving inverse problem to recover signal or images from degraded observations $\mathbf{y} = A\mathbf{x} + \boldsymbol{\epsilon}$,

- Traditionally solved by regularised optimisation $\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \frac{1}{2} \|A\mathbf{x} - \mathbf{y}\|_2^2 + \lambda \Phi(\mathbf{x})$,

- Motivation

- Classical iterative solvers (ISTA, FISTA, ADMM) suffer from:

- Slow convergence requiring hundreds to thousands of iterations.
 - Hand-crafted priors that may not capture complex image statistics.
 - Fixed step sizes and thresholds that are difficult to tune per dataset.

- Deep learning methods are black box

- Deep unfolding maps iterations to network layers

PROBLEM STATEMENT AND MOTIVATION

- Proximal Gradient method: Smooth + non-smooth parts

$$\mathbf{x}^{(k+1)} = \text{prox}_{\alpha\lambda\Phi} \left(\mathbf{x}^{(k)} - \alpha A^\top (A\mathbf{x}^{(k)} - \mathbf{y}) \right)$$

where $\alpha > 0$ is the step size and the proximal operator is:

$$\text{prox}_{\tau f}(\mathbf{v}) = \arg \min_{\mathbf{x}} f(\mathbf{x}) + \frac{1}{2\tau} \|\mathbf{x} - \mathbf{v}\|_2^2.$$

For $f = \|\cdot\|_1$, this reduces to soft-thresholding:

$$\mathcal{S}_\lambda(x) = \text{sign}(x) \cdot \max(|x| - \lambda, 0).$$

FISTA

Algorithm 1 FISTA for Sparse Recovery

Require: $\mathbf{y}$, A , λ , step size $\alpha = 1/L$, $L = \|A^\top A\|_2$

1: Initialise $\mathbf{x}^{(0)} = \mathbf{z}^{(1)} = \mathbf{0}$, $t_1 = 1$

2: **for** $k = 1, 2, \dots, K$ **do**

3: $\mathbf{x}^{(k)} \leftarrow \text{prox}_{\alpha\lambda\|\cdot\|_1} \left(\mathbf{z}^{(k)} - \alpha A^\top (A\mathbf{z}^{(k)} - \mathbf{y}) \right)$

4: $t_{k+1} \leftarrow \frac{1 + \sqrt{1 + 4t_k^2}}{2}$

5: $\mathbf{z}^{(k+1)} \leftarrow \mathbf{x}^{(k)} + \frac{t_k - 1}{t_{k+1}} \left(\mathbf{x}^{(k)} - \mathbf{x}^{(k-1)} \right)$

6: **end for** return $\mathbf{x}^{(K)}$

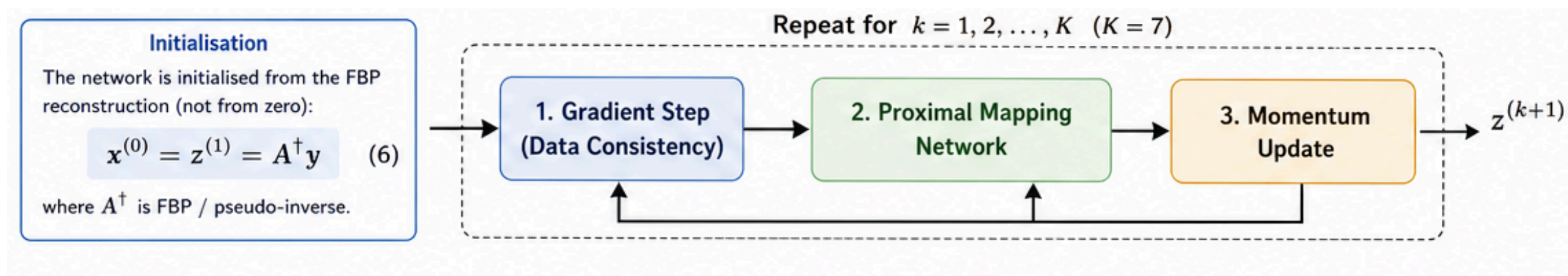
- FISTA accelerates proximal gradient descent with Nesterov momentum, achieving $O(1/k^2)$ convergence vs. $O(1/k)$ for standard ISTA:

DEEP UNFOLDING

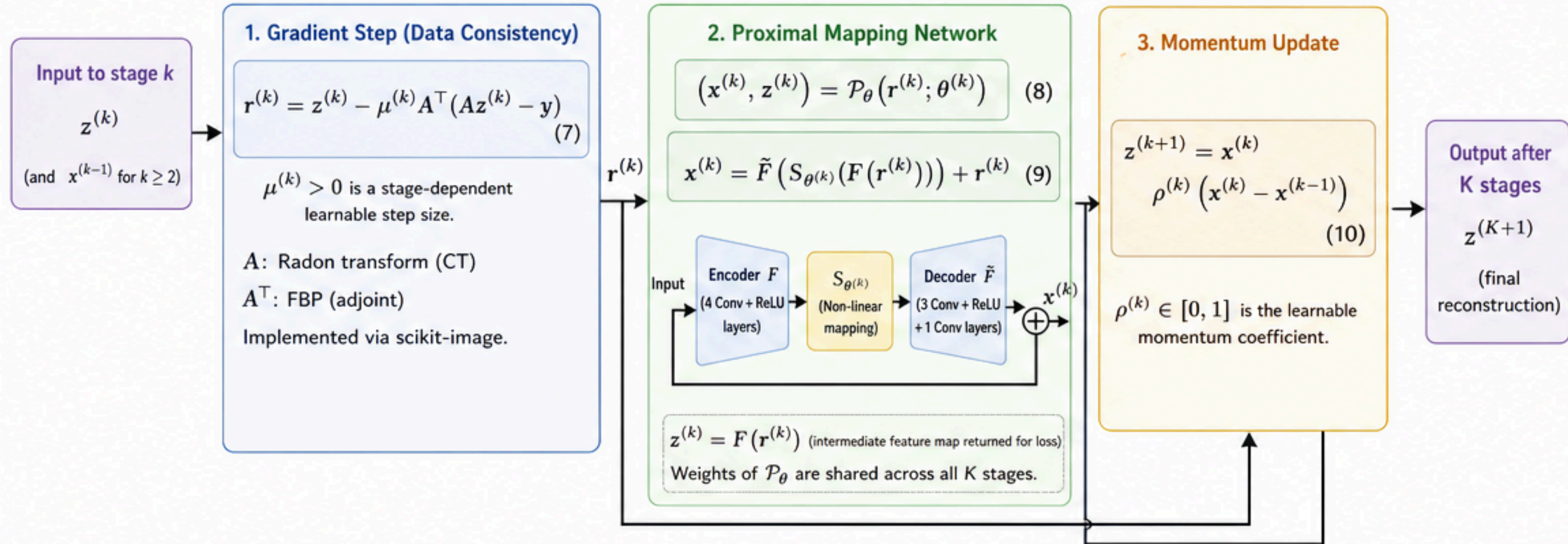
- LISTA- unfolding by mapping K ISTA iterations into a K -layer neural network with learnable weights.
- FISTA-Net extends this idea from ISTA to FISTA by introducing fully learnable step sizes, proximal thresholds, and momentum coefficients.
- Instead of using fixed optimization parameters, the entire reconstruction process is trained end-to-end for improved speed and accuracy.
- This enables faster convergence and better sparse signal recovery compared to classical iterative methods.

FISTA-NET ARCHITECTURE

- FISTA-NET unrolls $K=7$ FISTA iterations to K -staged network
- Each stage implements a gradient step, a convolutional proximal mapping, and a momentum update.
- Proximal network weights are shared across all stages
- Only the scalar parameters $\mu(k)$, $\theta(k)$, $\rho(k)$ vary per stage via learned smooth schedules parameterised by 6 scalars.



FISTA-NET ARCHITECTURE



The stage-wise parameters $\mu^{(k)}$, $\theta^{(k)}$, $\rho^{(k)}$ are smoothly parameterised using only 6 scalar parameters $\{w_1, c_1, w_2, c_2, w_3, c_3\}$:

Step size schedule

$$\mu^{(k)} = \text{softplus}(-\text{softplus}(-w_1) \cdot k + c_1) \quad (11)$$

Constraint: $\mu^{(k)} > 0$ (enforced by softplus)

Threshold schedule

$$\theta^{(k)} = \text{softplus}(-\text{softplus}(-w_2) \cdot k + c_2) \quad (12)$$

Constraint: $\theta^{(k)} \geq 0$ (enforced by softplus)

Momentum schedule

$$\rho^{(k)} = \text{clip}\left(\frac{\text{softplus}(w_3 k + c_3) - \text{softplus}(w_3 + c_3)}{\text{softplus}(w_3 k + c_3)}, 0, 1\right) \quad (13)$$

Constraint: $\rho^{(k)} \in [0, 1]$ (enforced by clip)

Initialisations: $w_1 = -0.5$, $c_1 = -2.0$, $w_2 = -0.2$, $c_2 = -1.0$, $w_3 = 1.0$, $c_3 = 0.0$.

EXPERIMENT DETAILS

Loss Function

The total training loss (Eq. 14 of the paper) combines three terms computed on the **final stage output only**:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{mse}} + \lambda_1 \mathcal{L}_{\text{sym}} + \lambda_2 \mathcal{L}_{\text{spa}},$$

with $\lambda_1 = 0.01$ and $\lambda_2 = 0.001$.

MSE loss (data fidelity, final stage only):

$$\mathcal{L}_{\text{mse}} = \left\| \mathbf{x}^{(K)} - \mathbf{x}^* \right\|_2^2$$

Symmetry loss (encourages $\tilde{F} \approx F^{-1}$, averaged over all stages):

$$\mathcal{L}_{\text{sym}} = \frac{1}{K} \sum_{k=1}^K \left\| \tilde{F} \left(F \left(\mathbf{x}^{(k)} \right) \right) - \mathbf{x}^{(k)} \right\|_2^2$$

Sparsity loss (promotes sparse transform-domain features, averaged over all stages):

$$\mathcal{L}_{\text{spa}} = \frac{1}{K} \sum_{k=1}^K \frac{1}{|\mathbf{z}^{(k)}|} \left\| \mathbf{z}^{(k)} \right\|_1$$

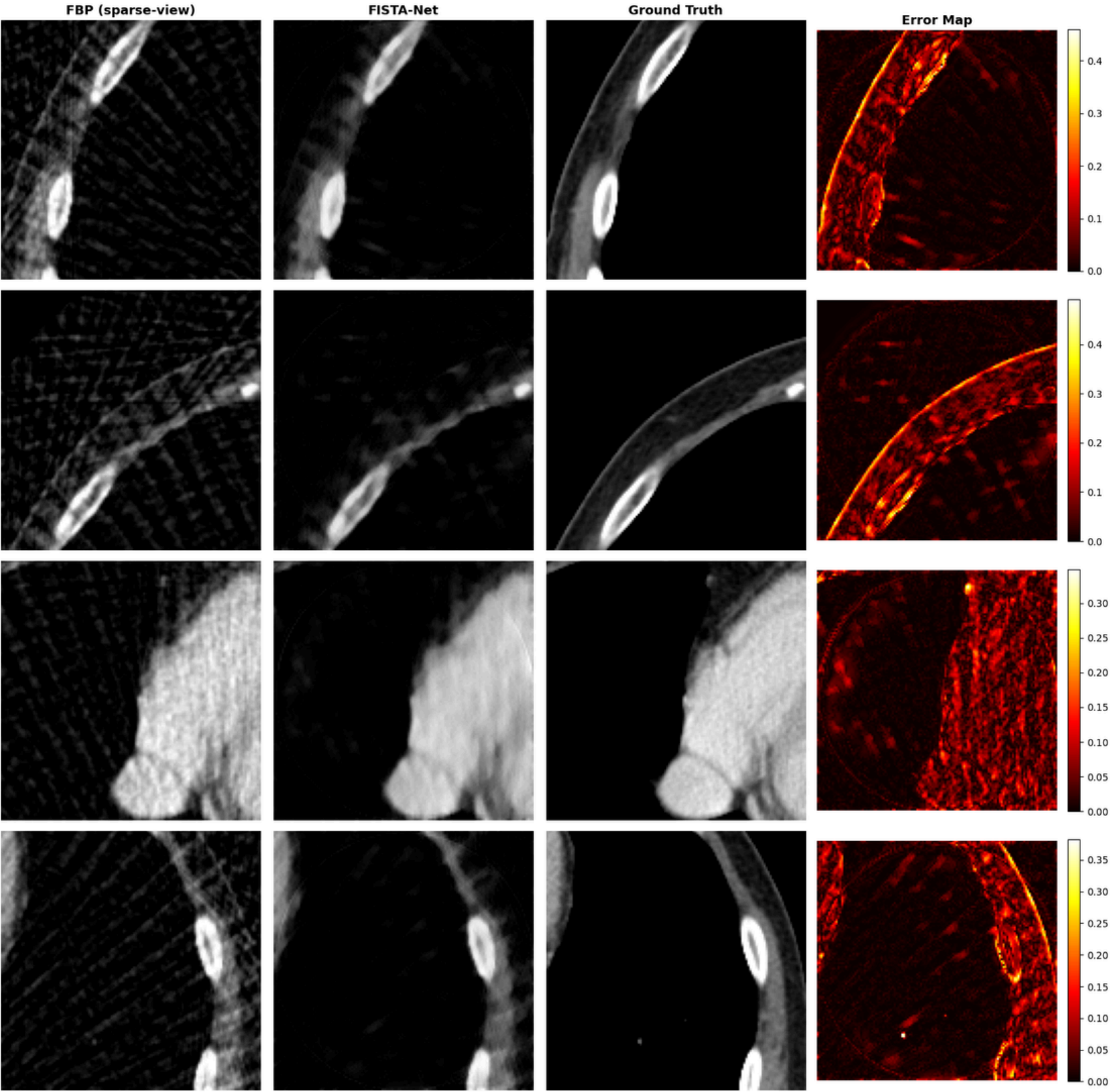
where $\mathbf{z}^{(k)} = F(\mathbf{r}^{(k)})$ are the encoder features before thresholding.

EXPERIMENT DETAILS

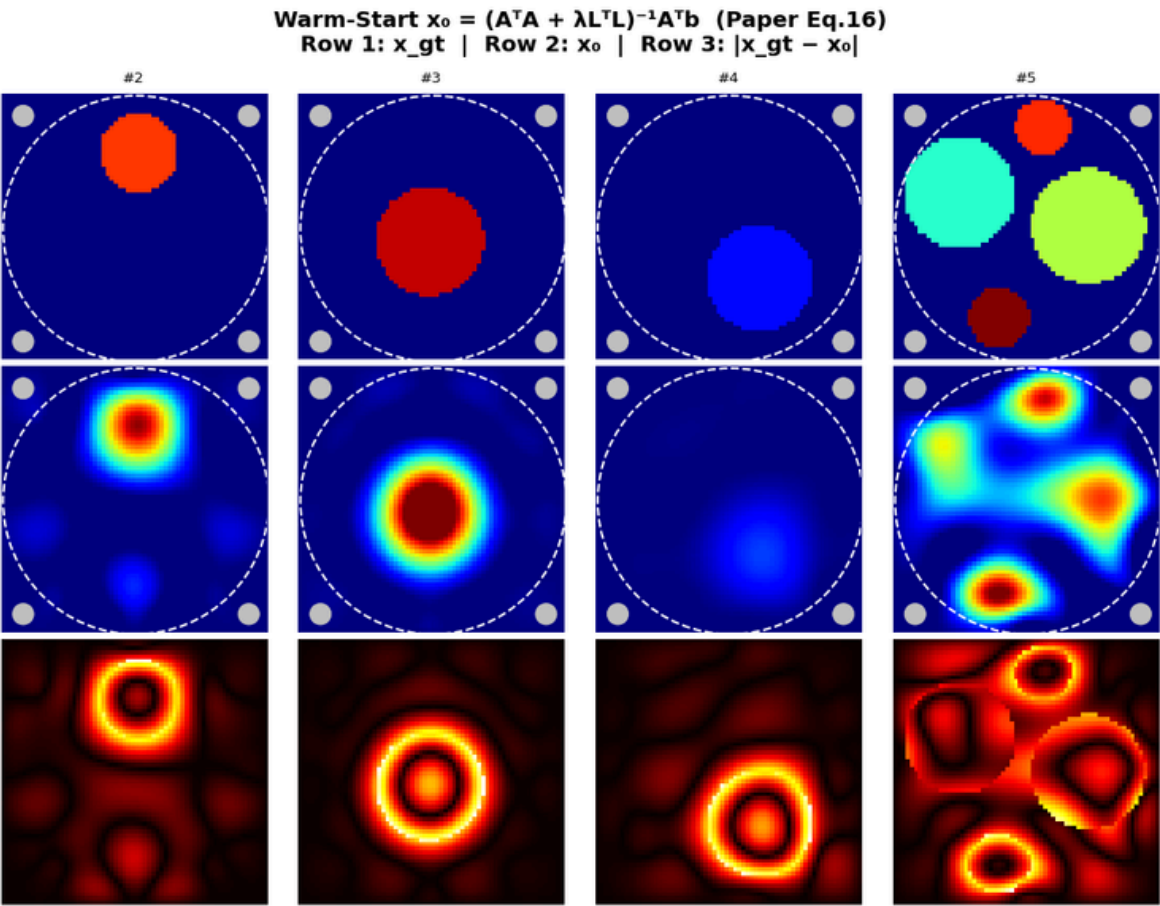
Parameters	CT	EMT
Epoch	10	50
Batch Size	2	16
Patch Size	128 X 128	64 X 64
LR	0.0001	0.0001
PARAMS	0.001	0.001
DATSET	100Mayo Clinic Sparse-View Dataset	Synthetic data

Hyperparameter	Value
Number of stages K	7
ProxNet filters	32
Total parameters (FISTA-Net)	55,878
Total parameters (FBPConvNet)	7,765,409
Init: w_1, c_1 (step size μ)	$-0.5, -2.0$
Init: w_2, c_2 (threshold θ)	$-0.2, -1.0$
Init: w_3, c_3 (momentum ρ)	1.0, 0.0
Optimiser	Adam
ProxNet lr	10^{-4}
Scalar params lr	10^{-3}
Gradient clip (max norm)	1.0
λ_1 (symmetry loss weight)	0.01
λ_2 (sparsity loss weight)	0.001
CT batch size	2
CT training epochs	10
CT patch size	128×128
CT training slices	240 (30/patient $\times$ 8)
CT val / test slices	10 / 10
HU window	$[-140, 260] \rightarrow [0, 1]$
Radon views	60
EMT epochs (smoke test)	1
EMT batch size	8

DATASET



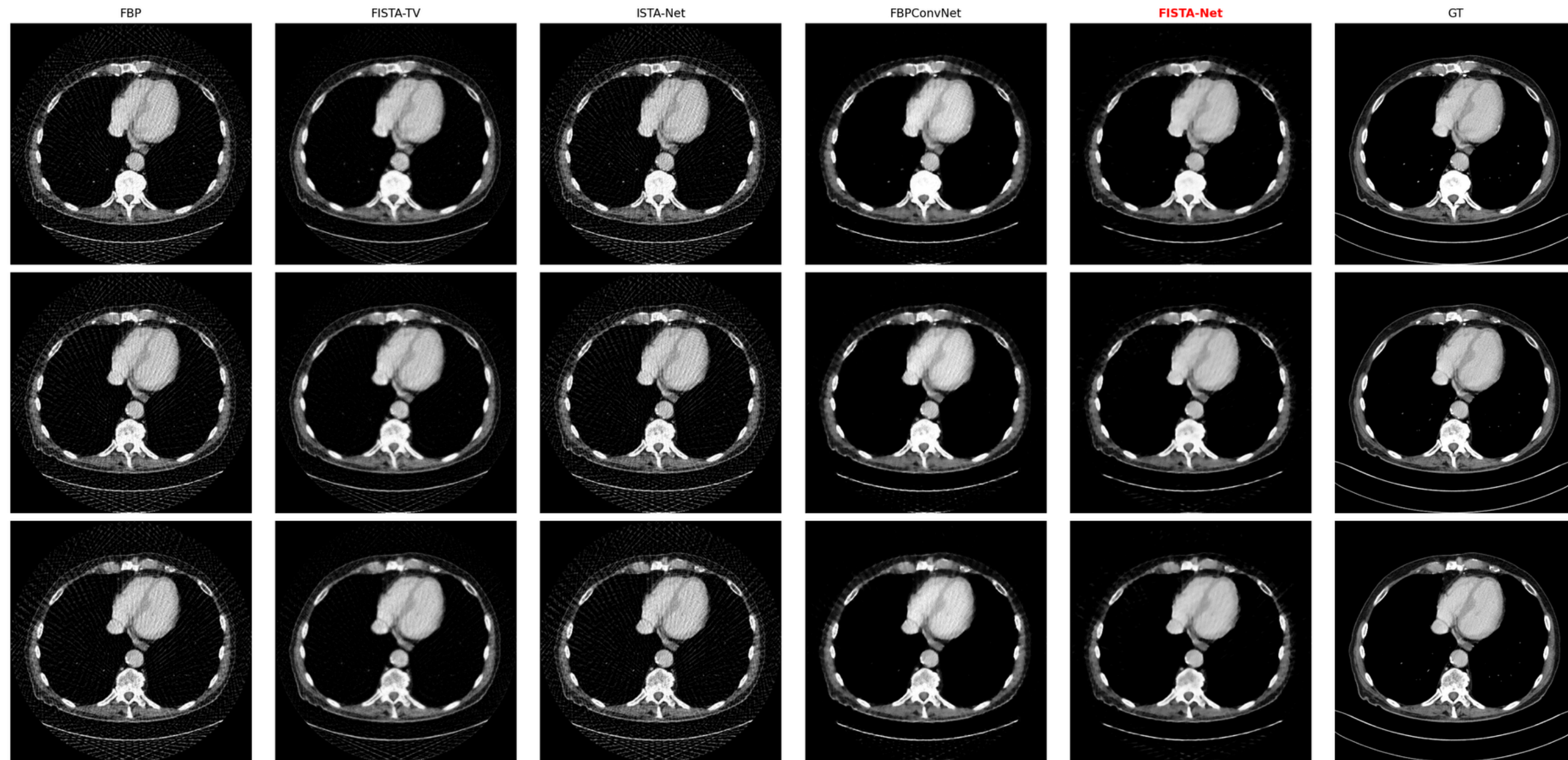
CT



EMT

RESULTS-CT

CT Reconstruction — All Methods (Mayo Clinic, 60-view)



- FISTA-Net produces reconstructions closest to the Ground Truth (GT) with clearer boundaries and better structural preservation.
- Compared to FISTA-TV and ISTA-Net, FISTA-Net shows fewer streak artifacts and sharper anatomical details.

RESULTS-CT

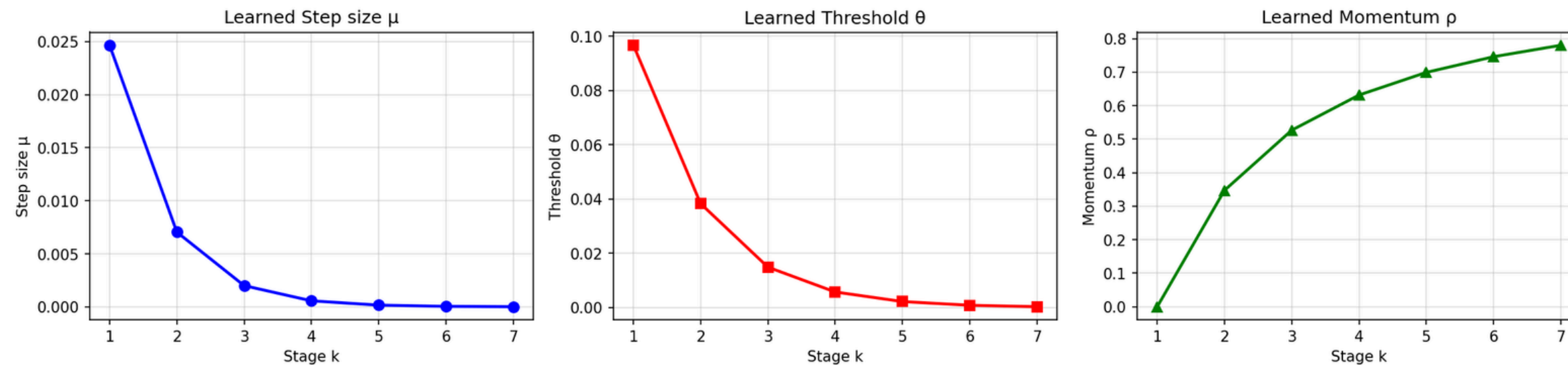
- All five methods were evaluated on the test patient L506 (10 full 512×512 slices, 60-view sparse acquisition)

Method	Type	PSNR (dB) $\uparrow$	SSIM $\uparrow$	RMSE $\downarrow$
FBP	Classical	21.62 ± 0.19	0.425 ± 0.008	0.0830 ± 0.0018
FISTA-TV	Classical	23.30 ± 0.19	0.530 ± 0.008	0.0684 ± 0.0015
ISTA-Net [†]	Deep (unrolled)	21.59 ± 0.19	0.424 ± 0.008	0.0833 ± 0.0018
FBPConvNet	Deep (post-proc.)	23.86 ± 0.25	0.553 ± 0.005	0.0642 ± 0.0019
FISTA-Net (ours)	Deep (unrolled)	23.80 ± 0.16	0.764 ± 0.002	0.0646 ± 0.0012

- FISTA-Net and FBPConvNet show nearly identical performance, with FBPConvNet leading by only 0.06 dB PSNR and 0.0004 RMSE.
- FISTA-Net outperforms FISTA-TV in PSNR and SSIM, proving learned parameters are better than hand-tuned classical optimization.
- FISTA-Net has lower PSNR variance than FBPConvNet, indicating more consistent reconstruction quality across patient slices.

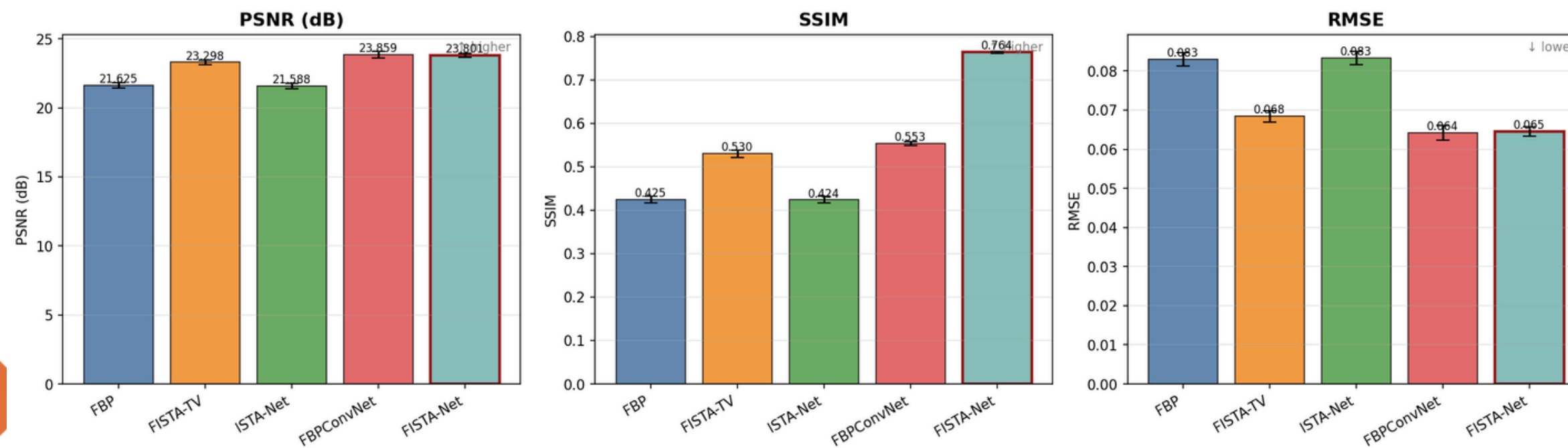
RESULTS-CT

FISTA-Net Learned Parameter Schedules (CT)



- μ and θ decrease with stages, giving finer reconstruction updates.
- ρ increases with stages, helping faster and more stable convergence.

CT Reconstruction Metrics — Mayo Clinic 60-view Sparse CT



- FISTA-Net performs very close to FBPCovNet in PSNR and RMSE.
- FISTA-Net clearly outperforms classical FISTA-TV in PSNR and SSIM.
- Lower variance shows more consistent reconstruction quality across slices.

RESULTS-EMT

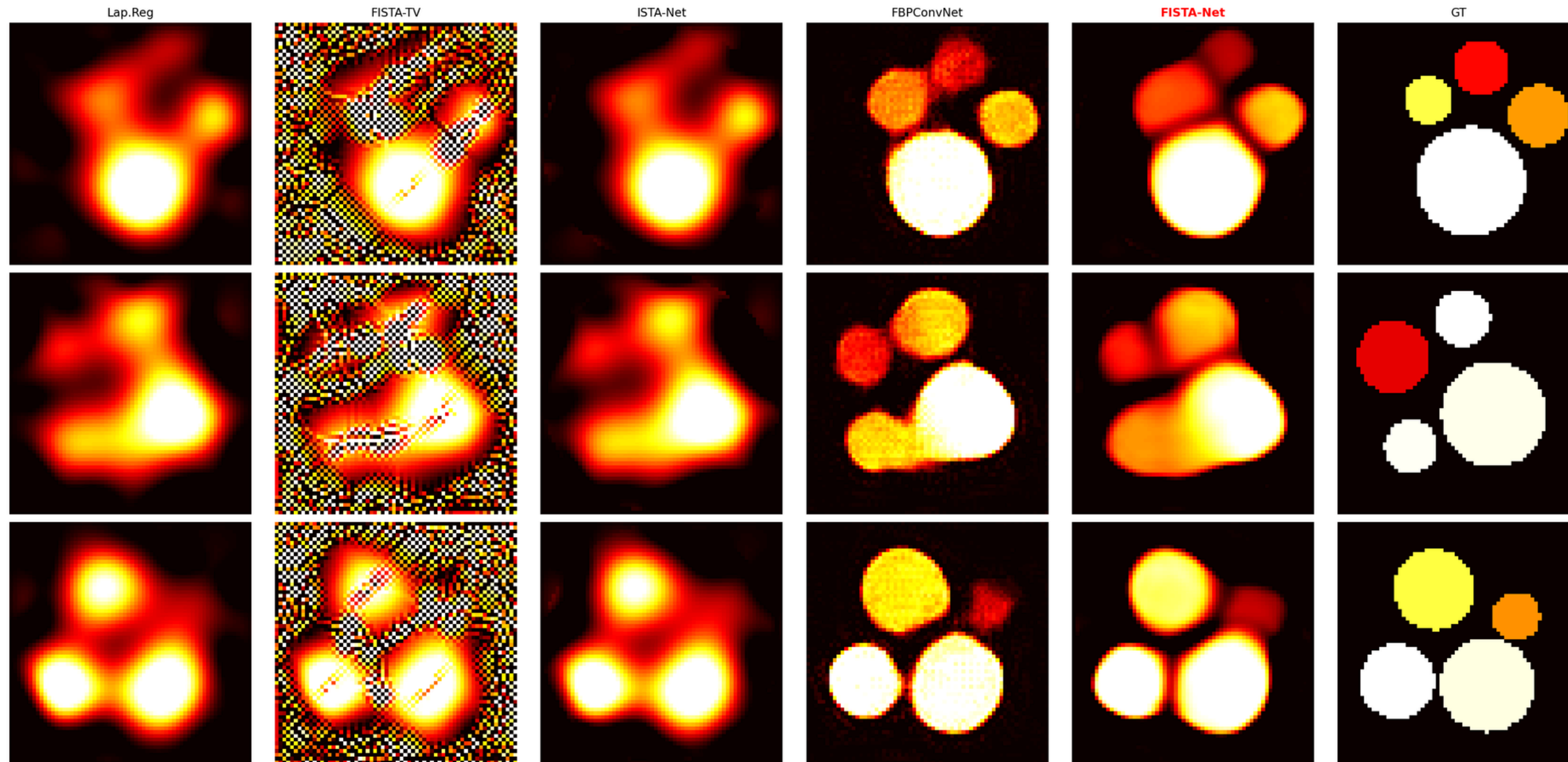
Method	Type	PSNR (dB) $\uparrow$	SSIM $\uparrow$	RMSE $\downarrow$
Lap. Reg.	Classical	16.39 ± 1.62	0.249 ± 0.048	0.0551 ± 0.0228
FISTA-TV	Classical	$0.75 \pm 4.53^\dagger$	0.061 ± 0.034	0.2998 ± 0.0302
ISTA-Net	Deep (unrolled)	16.83 ± 1.76	0.529 ± 0.099	0.0526 ± 0.0222
FBPConvNet	Deep (post-proc.)	20.28 ± 3.34	0.663 ± 0.102	0.0379 ± 0.0214
FISTA-Net (ours)	Deep (unrolled)	18.99 ± 2.56	0.754 ± 0.106	0.0422 ± 0.0206

† FISTA-TV diverged on this synthetic measurement matrix (near-zero PSNR).

- FISTA-TV diverges on random Gaussian measurement matrices because the TV proximal step is highly sensitive to step-size tuning
- FBPConvNet achieves the best PSNR (20.28 dB) and RMSE (0.0379) due to its larger capacity (7.77M parameters) and simpler MSE-only training objective.
- FISTA-Net achieves the best SSIM (0.754), outperforming FBPConvNet and ISTA-Net, stronger preservation of structural and perceptual image details.
- FISTA-Net outperforms ISTA-Net on PSNR, SSIM, and RMSE, confirming that momentum acceleration significantly improves reconstruction quality and convergence speed.

RESULTS-EMT

EMT Reconstruction — All Methods



- FISTA-Net achieves reconstructions closest to the Ground Truth (GT) with clearer object boundaries and better shape preservation.
- Compared to ISTA-Net and FISTA-TV, FISTA-Net shows fewer distortions and significantly reduced reconstruction artifacts.

LIMITATION

- Training epochs: 10 epochs on 240 slices is far from the paper's training regime;
- results are expected to improve substantially with more compute.
- Patch-based training: 128×128 crops may miss global structures; full 512×512 training would better replicate the paper.
- CPU Radon operator: Limits batch size to 2 and slows training significantly.

CONCLUSION

- We implemented and evaluated FISTA-Net for two inverse problems: sparse-view CT reconstruction and synthetic EIT/EMT reconstruction.
- In CT reconstruction, FISTA-Net achieved the best SSIM (0.764) while matching FBPCNN in PSNR and RMSE.
- It reached similar performance using 139× fewer parameters (55,878 vs 7.77M), showing high efficiency.
- FISTA-Net also outperformed FISTA-TV on all metrics, proving learned optimization is better than hand-tuned iterative methods.
- In EMT reconstruction, FISTA-Net achieved the best SSIM (0.754), preserving structural details better than other methods.
- FBPCNN achieved the best PSNR and RMSE due to larger model capacity and simpler training.
- FISTA-Net outperformed ISTA-Net on all three metrics, showing the benefit of learned momentum.
- FISTA-TV diverged on random Gaussian measurements, showing the instability of classical methods under ill-conditioned operators.

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THANK YOU